Then along came quantum mechanics, and everything changed.

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Alongside Newton's formulation of classical mechanics, the invention of quantum mechanics represents the other great revolution in the history of physics. Unlike anything that had come before, quantum theory didn't propose a particular physical model within the basic classical framework; it discarded that framework entirely, replacing it with something profoundly different.

The fundamental new element of quantum mechanics, the thing that makes it unequivocally distinct from its classical predecessor, centers on the question of what it means to *measure* something about a quantum system. What exactly a measurement is, and what happens when we measure something, and what this all tells us about what's really happening behind the scenes: together, these questions constitute what's called the *measurement problem* of quantum mechanics. There is absolutely no consensus within physics or philosophy on how to solve the measurement problem, although there are a number of promising ideas.

Attempts to address the measurement problem have led to the emergence of a field known as *the interpretation of quantum mechanics*, although the label isn't very accurate. "Interpretations" are things that we might apply to a work of literature or art, where people might have different ways of thinking about the same basic object. What's going on in quantum mechanics is something else: a competition between truly distinct scientific theories, incompatible ways of making sense of the physical world. For this reason, modern workers in this field prefer to call it "foundations of quantum mechanics." The subject of quantum foundations is part of science, not literary criticism.

Nobody ever felt the need to talk about "interpretations of classical mechanics"—classical mechanics is perfectly transparent. There is a mathematical formalism that speaks of positions and velocities and trajectories, and oh, look: there is a rock whose actual motion in the world obeys the predictions of that formalism. There is, in particular, no such thing as a measurement problem in classical mechanics. The state of the system is given by its position and its velocity, and if we want to measure